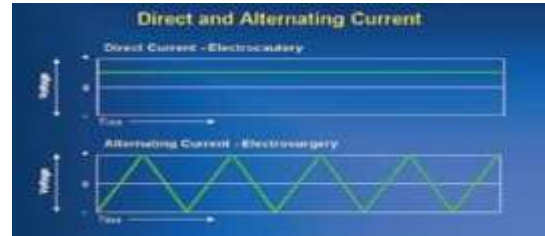
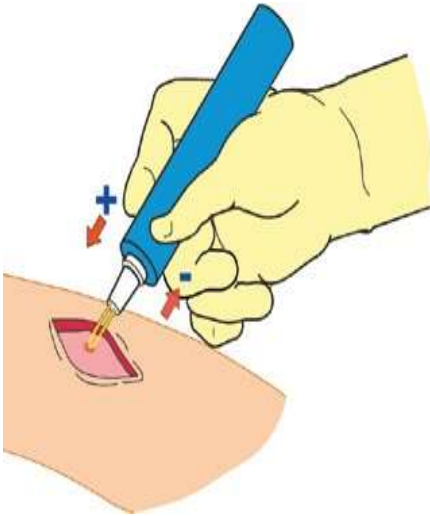


# Principles of electrosurgery

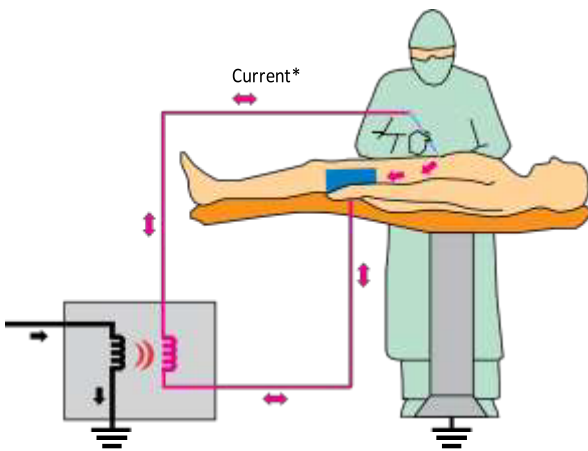
## Electrocautery

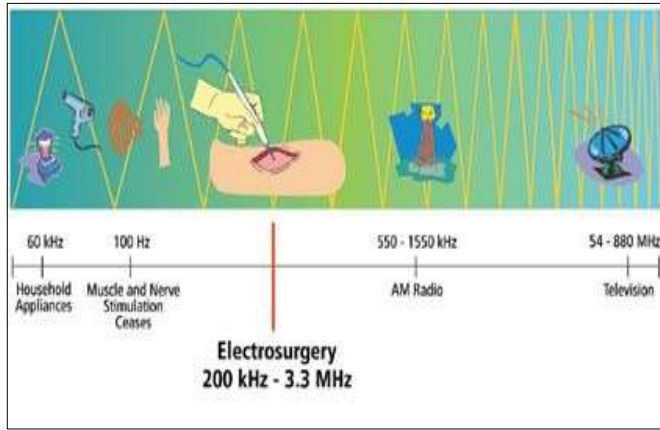
Often “electrocautery” is used to describe electrosurgery. This is incorrect. Electrocautery refers to direct current (electrons flowing in one direction) whereas electrosurgery uses alternating current. During electrocautery, current does not enter the patient’s body. Only the heated wire comes in contact with tissue. In electrosurgery, the patient is included in the circuit and current enters the patient’s body



## Principles of Electrosurgery in the O.r.

Principles of electricity are relevant in the operating room. The electrosurgical generator is the source of the electron flow and voltage. The circuit is composed of the generator, active electrode, patient and patient return electrode. Pathways to ground are numerous but may include the O.R. table, stirrups, staff members and equipment. The patient’s tissue provides the impedance, producing heat as the electrons overcome the impedance.





## frequency Spectrum

Standard electrical current alternates at a frequency of 60 cycles per second (Hz). Electrosurgical systems could function at this frequency, but because current would be transmitted through body tissue at 60 cycles, excessive neuromuscular stimulation and perhaps electrocution would result.

Because nerve and muscle stimulation cease at 100,000 cycles/ second (100 kHz), electrosurgery can be performed safely at “radio” frequencies above 100 kHz. An electrosurgical generator takes 60 cycle current and increases the frequency to over 200,000 cycles per second. At this frequency electrosurgical energy can pass through the patient with minimal neuromuscular stimulation and no risk of electrocution.

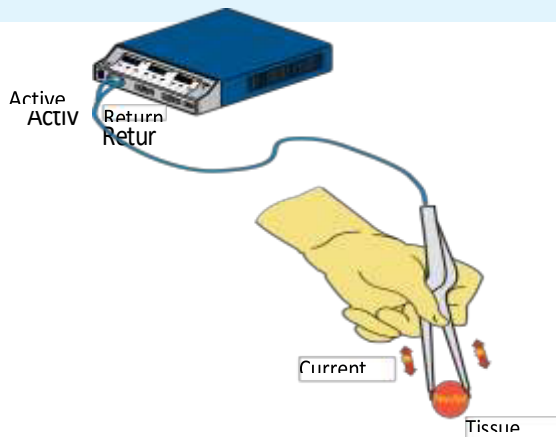
- **Active output and patient return functions are both accomplished at the site of surgery.**
- **Current path is confined to tissue grasped between forceps tines.**
- **Patient return electrode should not be applied for bipolar only procedures.**

## Bipolar electrosurgery

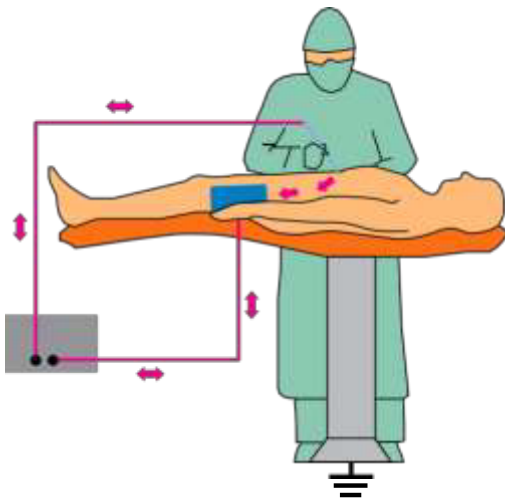
In bipolar electrosurgery, both the active electrode and return electrode functions are performed at the site of surgery. The two tines of the forceps perform the active and return electrode functions. Only the tissue grasped is included in the electrical circuit. Because the return function is performed by one tine of the forceps, no patient return electrode is needed.

## Bipolar Circuit

This picture represents a typical bipolar circuit.



- The active electrode is in the wound.
- The patient return electrode is attached somewhere else on the patient.
- The current must flow through the patient to the patient return electrode.



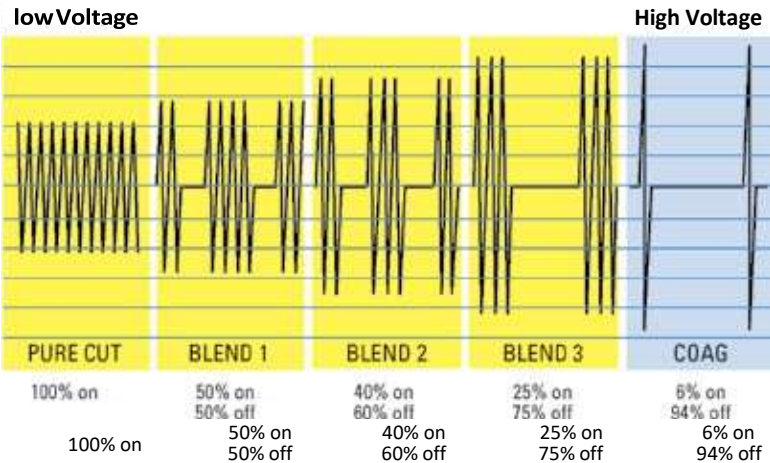
## Monopolar electrosurgery

Monopolar is the most commonly used electrosurgical modality. This is due to its versatility and clinical effectiveness. In monopolar electrosurgery, the active electrode is in the surgical site. The patient return electrode is somewhere else on the patient's body. The current passes through the patient as it completes the circuit from the active electrode to the patient return electrode.

## Monopolar Circuit

This picture represents a common monopolar circuit. There are four components to the monopolar circuit:

- generator
- Active Electrode
- Patient
- Patient return Electrode



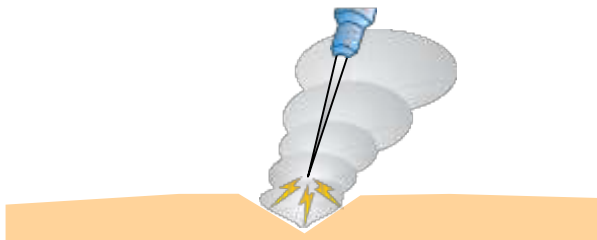
## Tissue effects change as you modify the wave form

Electrosurgical generators are able to produce a variety of electrical waveforms. As waveforms change, so will the corresponding tissue effects. Using a constant waveform, like “cut,” the surgeon is able to vaporize or cut tissue. This waveform produces heat very rapidly.

Using an intermittent waveform, like “coagulation,” causes the generator to modify the waveform so that the duty cycle (“on” time) is reduced. This interrupted waveform will produce less heat. Instead of tissue vaporization, a coagulum is produced.

A “blended current” is not a mixture of both cutting and coagulation current but rather a modification of the duty cycle. As you go from Blend 1 to Blend 3 the duty cycle is progressively reduced. A lower duty cycle produces less heat. Consequently, Blend 1 is able to vaporize tissue with minimal hemostasis whereas Blend 3 is less effective at cutting but has maximum hemostasis.

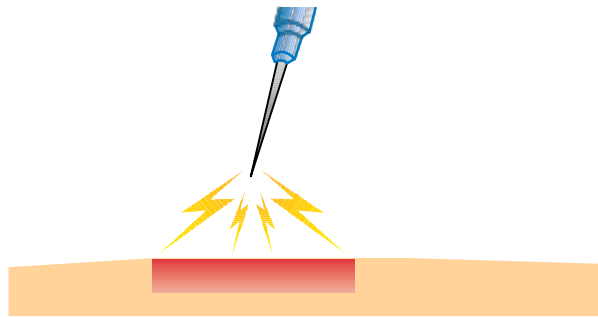
The only variable that determines whether one waveform vaporizes tissue and another produces a coagulum is the rate at which heat is produced. High heat produced rapidly causes vaporization. Low heat produced more slowly creates a coagulum. Any one of the five waveforms can accomplish both tasks by modifying the variables that impact tissue effect.



## Electrosurgical tissue effects

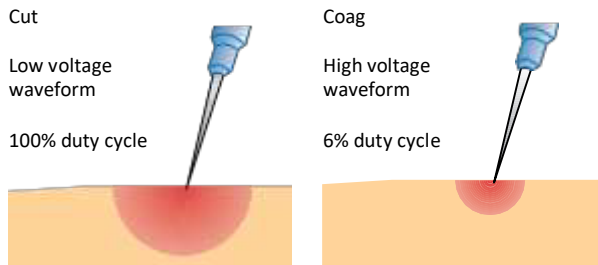
### Electrosurgical Cutting

Electrosurgical cutting divides tissue with electric sparks that focus intense heat at the surgical site. By sparking to tissue, the surgeon produces maximum current concentration. To create this spark the surgeon should hold the electrode slightly away from the tissue. This will produce the greatest amount of heat over a very short period of time, which results in vaporization of tissue.



## fulguration

Electrosurgical fulguration (sparking with the coagulation waveform) coagulates and chars the tissue over a wide area. Since the duty cycle (on time) is only about 6 percent, less heat is produced. The result is the creation of a coagulum rather than cellular vaporization. In order to overcome the high impedance of air, the coagulation waveform has significantly higher voltage than the cutting current. Use of high voltage coagulation current has implications during minimally invasive surgery.

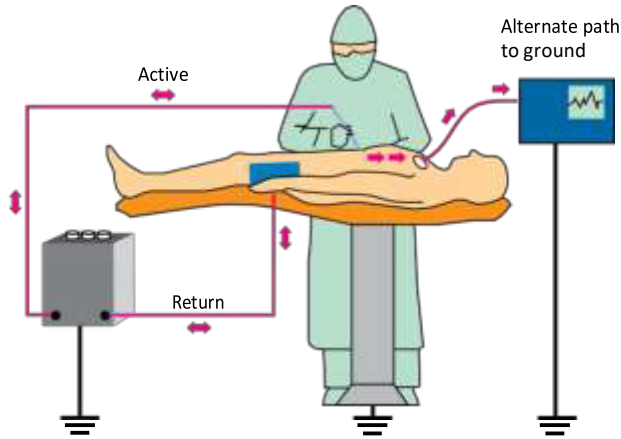


## Desiccation

Electrosurgical desiccation occurs when the electrode is in direct contact with the tissue. Desiccation is achieved most efficiently with the “cutting” current. By touching the tissue with the electrode, the current concentration is reduced. Less heat is generated and no cutting action occurs. The cells dry out and form a coagulum rather than vaporize and explode.

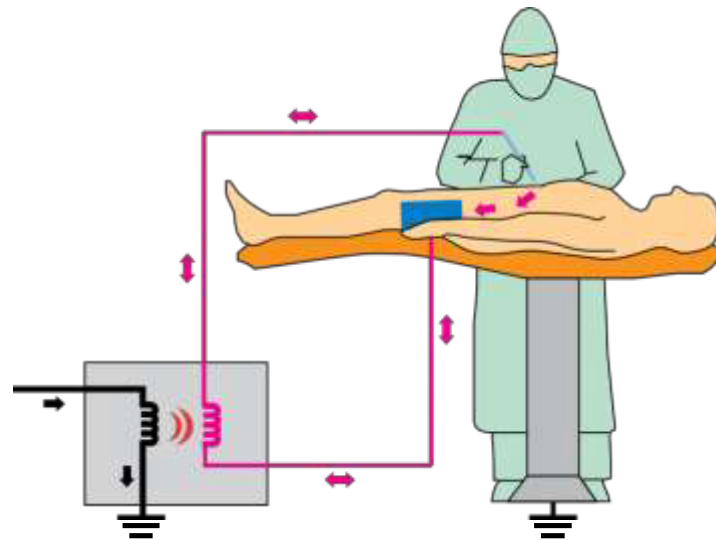
Many surgeons routinely “cut” with the coagulation current. Likewise, you can coagulate with the cutting current by holding the electrode in direct contact with tissue. It may be necessary to adjust power settings and electrode size to achieve the desired surgical effect. The benefit of coagulating with the cutting current is that you will be using far less voltage. Likewise, cutting with the cut current will also accomplish the task with less voltage. This is an important consideration during minimally invasive procedures.





### Grounded electro-surgical systems.

Originally, generators used grounded current from a wall outlet. It was assumed that once the current entered the patient's body, it would return to ground through the patient return electrode. But electricity will always seek the path of least resistance. When there are many conductive objects touching the patient and leading to ground, the current will select as its pathway to ground the most conductive object – which may not be the patient return electrode. Current concentration at this point may lead to an alternate site burn.



### Isolated electro-surgical systems

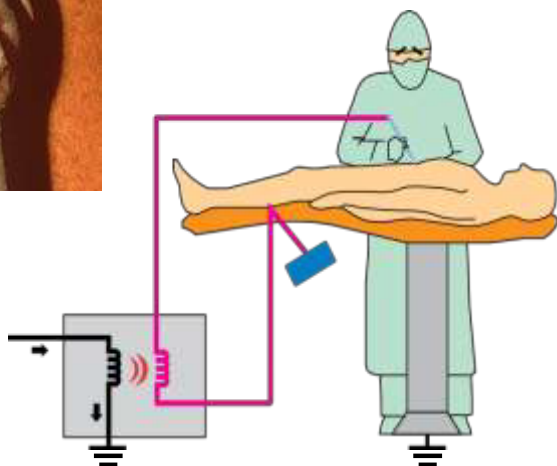
#### Isolated System

In 1968, electrosurgery was revolutionized by isolated generator technology. The isolated generator isolates the therapeutic current from ground by referencing it within the generator circuitry. In other words, in an isolated electro-surgical system, the circuit is completed not by the ground but by the generator. Even though grounded objects remain in the operating room, electro-surgical current from isolated generators will not recognize grounded objects as pathways to complete the circuit. Isolated electro-surgical energy recognizes the patient return electrode as the preferred pathway back to the generator.

By removing ground as a reference for the current, the isolated generator eliminates many of the hazards inherent in grounded systems, most importantly current division and alternate site burns.



Pad site



## Deactivated Isolated System

If the circuit to the patient return electrode is broken, an isolated generator will deactivate the system because the current cannot return to its source.

Generators with isolated circuits mitigate the hazard of alternate site burns but do not protect the patient from return electrode burns, such as the one shown at left.

Historically, patient return electrode burns have accounted for 70 percent of the injuries reported during the use of electrosurgery. Patient return electrodes are not “inactive” or “passive.” The only difference between the “active” electrode and the patient return electrode is their size and relative conductivity. The quality of the conductivity and contact area at the pad/patient interface must be maintained to prevent a return electrode site injury.

## patient return Electrode

### function of the Patient return Electrode

The function of the patient return electrode is to remove current from the patient safely.

A return electrode burn occurs when the heat produced, over time, is not safely dissipated by the size or conductivity of the patient return electrode.

Current \* time

Burn = -----

Area





COVIDIEN